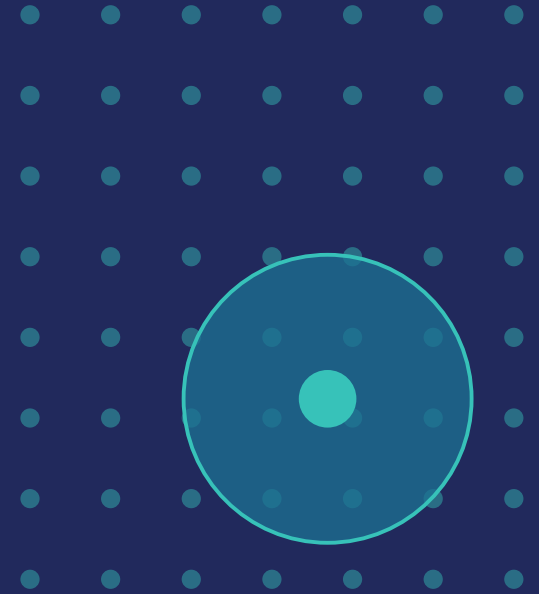


INTRODUCTION TO

# Data Science

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# What's so important about data?

---



# The Big Three

The three teams that sit around the data in most organisations

1

## Data Team

Solves a business problem and does a significant amount of work on the data that is available first.

2

## Business Intelligence Team

Provides business insights through dashboards and reporting.

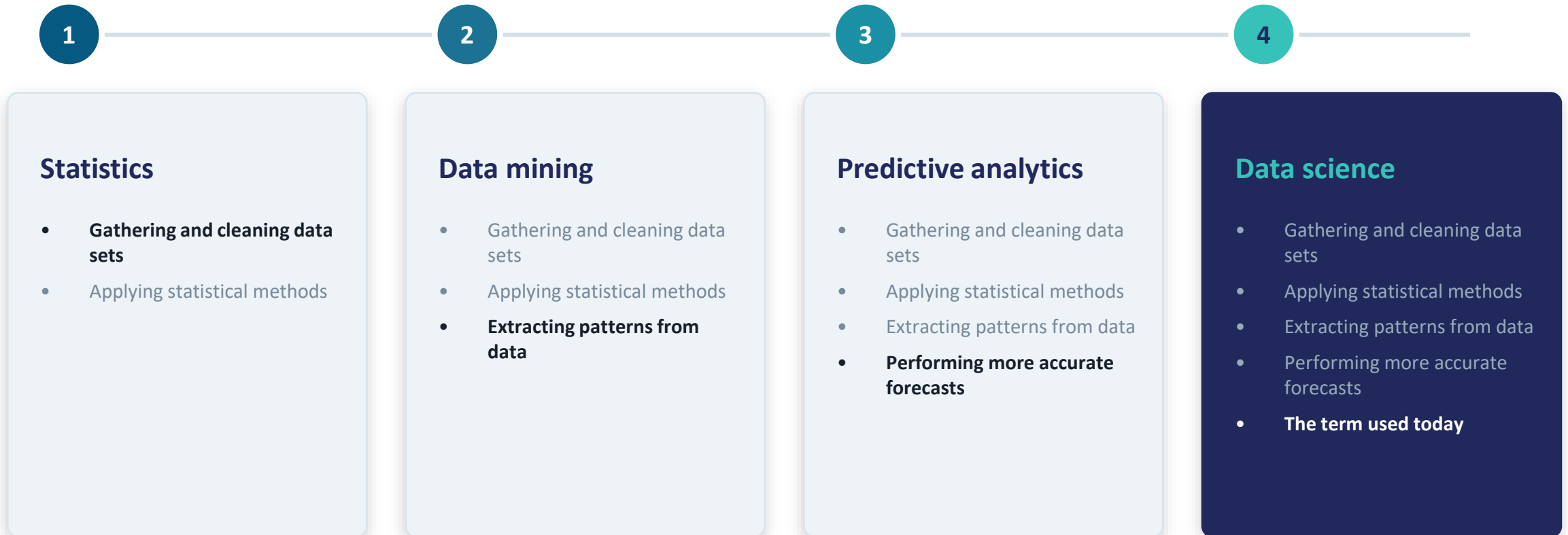
3

## Data Science Team

Uses business analytics or data analytics tools to build models that predict future outcomes.

# Evolution of the data science industry

Each stage keeps everything before it and adds one new capability



*Driven by the growth of data and radical improvements in technology*

# Data Analysis vs Data Analytics

Two words people use interchangeably. They are not the same thing.



?

**What is Data Analysis?**

?

**What is Data Analytics?**

# Analysis

1

## Break it down

Divide a dataset into manageable chunks, study them individually, then examine how they relate to the other parts.

2

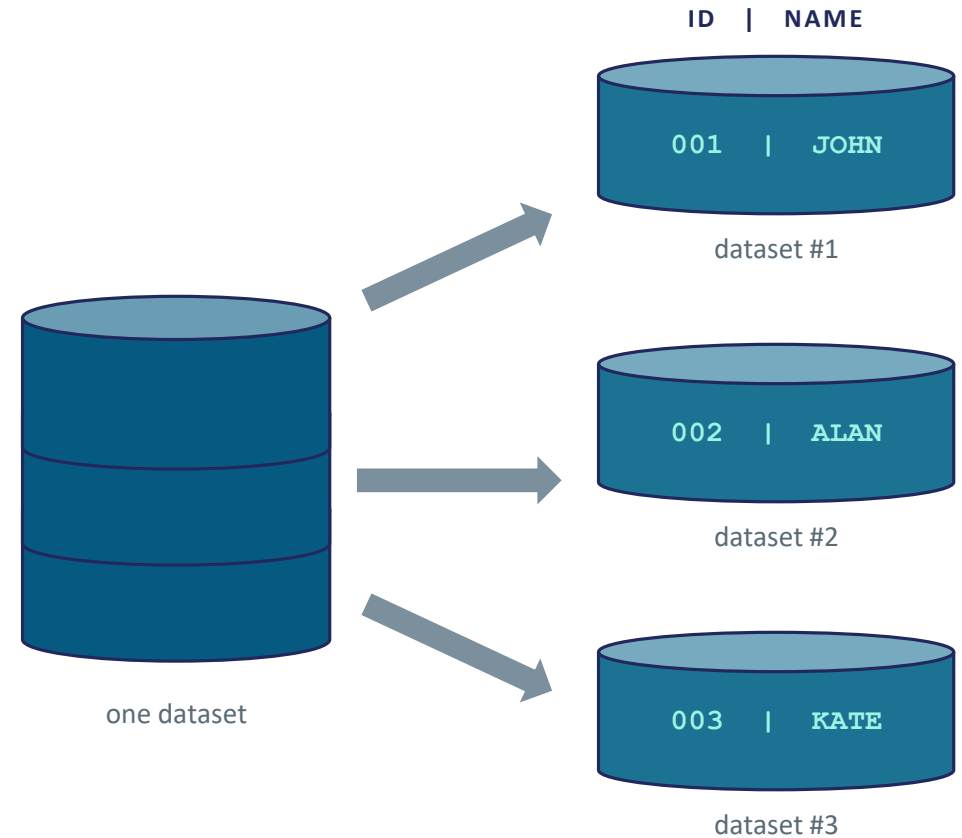
## Look at what happened

You are working with events that have already taken place.

3

## Answer How and Why

The output is an explanation, not a prediction.



# FUTURE

Analytics explores what could happen  
next



## Analytics

1

### Explore what's ahead

Analytics is aimed at potential future events.

2

### Apply reasoning

Logical and computational reasoning applied to the component parts you obtained in an analysis.

3

### Turn patterns into action

Look for patterns, then explore what you could do with them going forward.

# Qualitative & Quantitative Analytics

Two ways of getting to an answer

## Qualitative

### Intuition + Analysis

Human judgment applied to what the data shows.

## Quantitative

### Formulas + Algorithms

Structured methods that scale beyond what a person can hold in mind.





## KNOWLEDGE CHECK

**Which of the following is associated with the meaning of the term "analytics"?**

- A** Dividing a dataset into manageable chunks, studying them individually, and examining their relationships with other parts
- B** Explaining past events through analytics
- C** Essentially, analytics involves applying logical and computational reasoning to the components derived from an analysis
- D** None of the above

# Business Analytics



Analytics applied specifically to business data, aimed at a decision rather than at discovery for its own sake.

## ● What it works with

Sales, operations, marketing, and finance data. The everyday records a business already generates.

## ● What it asks

How are we performing, what is driving it, and what is likely to happen next quarter.

## ● What it produces

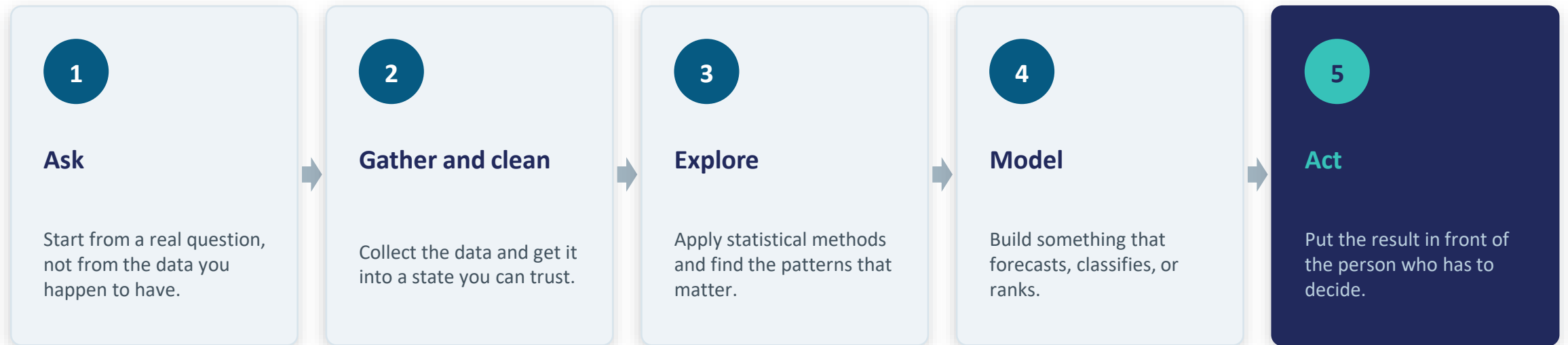
Dashboards, reports, and forecasting models that a manager can actually act on.

## ● Where it sits

The bridge between the BI team's reporting and the data science team's predictive models.

# Data Science

Data science is a discipline reliant on data availability, while business analytics does not completely rely on data. It is mostly the part that uses complex mathematical, statistical, and programming tools.



*Every earlier stage still lives inside it. Data science is simply the name the industry settled on.*

# Business Intelligence (BI)

The process of analysing and reporting historical business data.

**Aims to explain past events using business data**

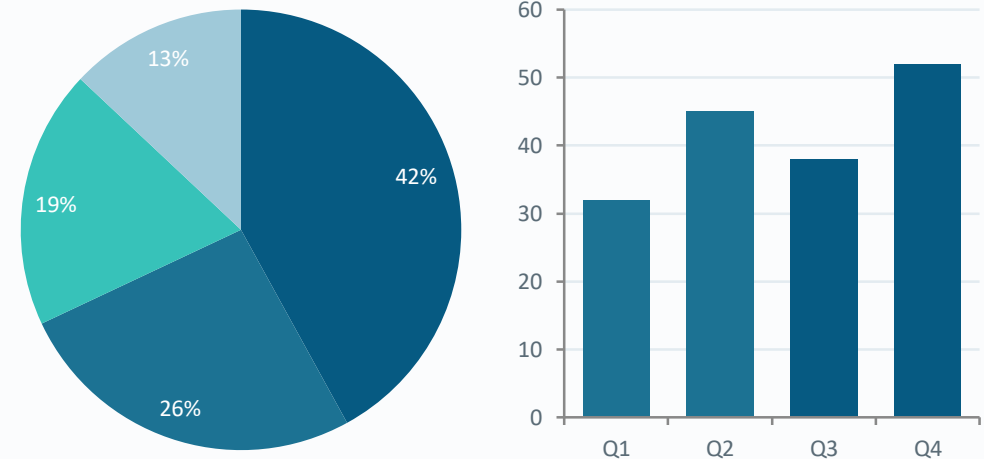
## Backward looking

It reports what already happened, it does not forecast.

## Shared view

Everyone reads performance from the same numbers.

**BI Dashboard**

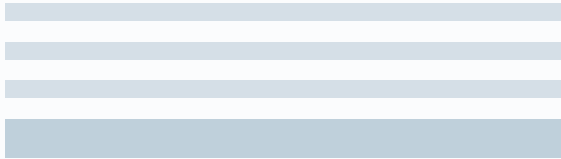


*Historical performance, read at a glance*

# Business Intelligence in Practice

Includes all technology-driven tools involved in the process of **analyzing**, **understanding** and **reporting** available past data.

## BI Report



## Make decisions

On evidence, rather than on instinct or the loudest voice.



## Extract insights

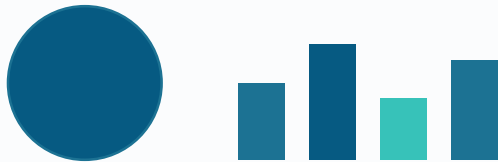
Work out what the numbers are actually saying.



## Extract ideas

Surface new questions worth chasing next.

## BI Dashboard



# Machine Learning (ML)



The ability of machines to predict outcomes without being explicitly programmed.

**ML is about creating and implementing algorithms that let machines receive data and use this data to:**



## Make predictions

Estimate what is likely to happen next, from data it has already seen.



## Analyse patterns

Find structure in volumes of data far too large to eyeball.

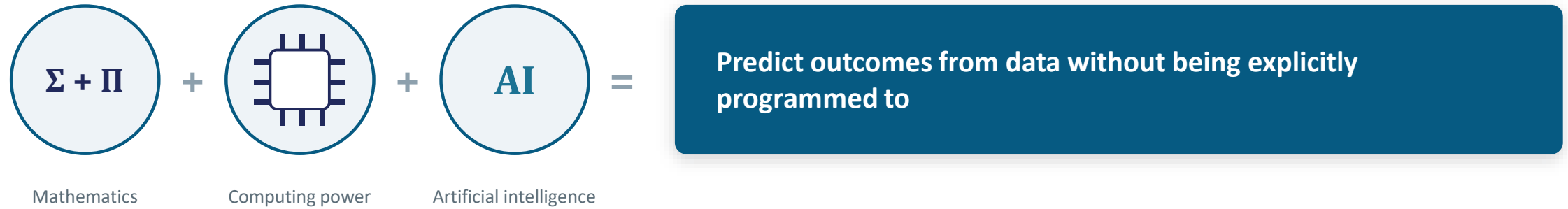


## Give recommendations

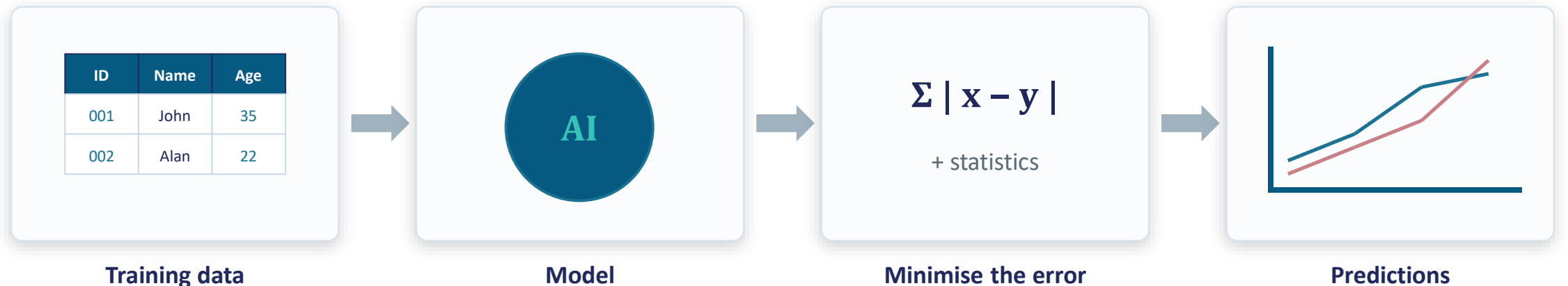
Suggest the next action, product, or route worth taking.

*Machine learning is the engine behind the predictive side of data science.*

# How Machine Learning Works



## The algorithm, step by step



*Predictions with unprecedented accuracy*

# Artificial Intelligence (AI)

Simulating human knowledge and decision making with computers.

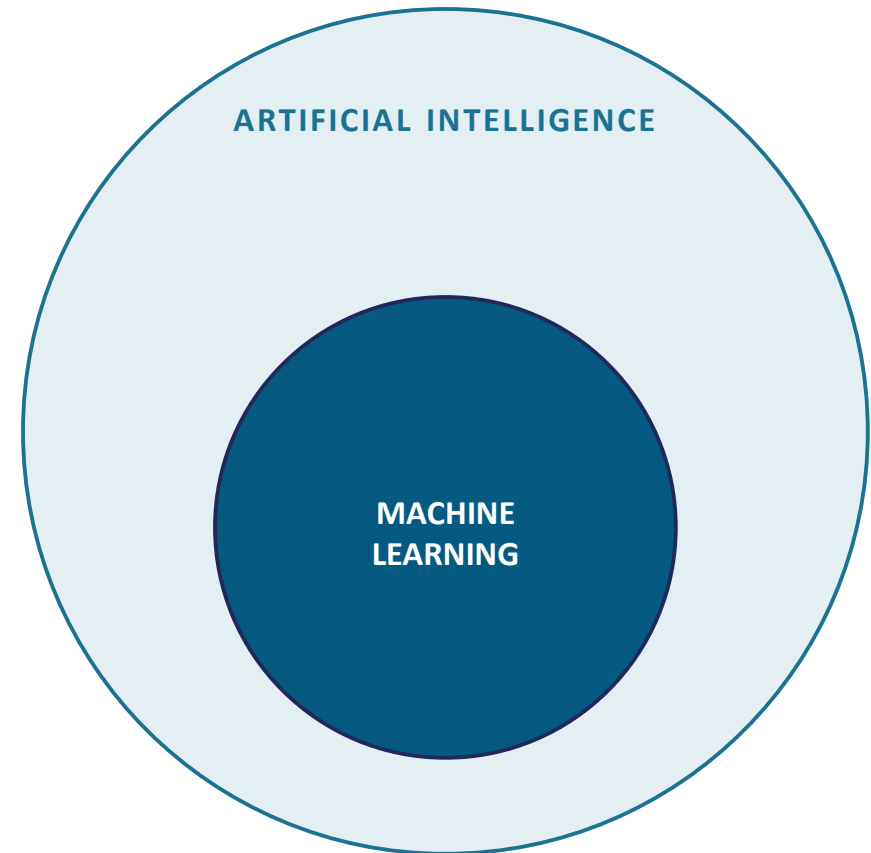
**We, as humans, have only managed to reach AI through machine learning**

- **Judgment, not just calculation**

The aim is decisions a person would otherwise have to make.

- **Learned, not hand-written**

The rules come from data, which is why ML is the route in.



*Machine learning is the part of AI we have actually built.*



# Traditional AI vs Generative AI

Two different jobs, both sitting under the same AI umbrella.

## TRADITIONAL AI

Mainly applied to structured datasets to improve efficiency and accuracy in repetitive tasks.

*Forecasting, fraud detection, recommendation ranking.*

VS

## GENERATIVE AI

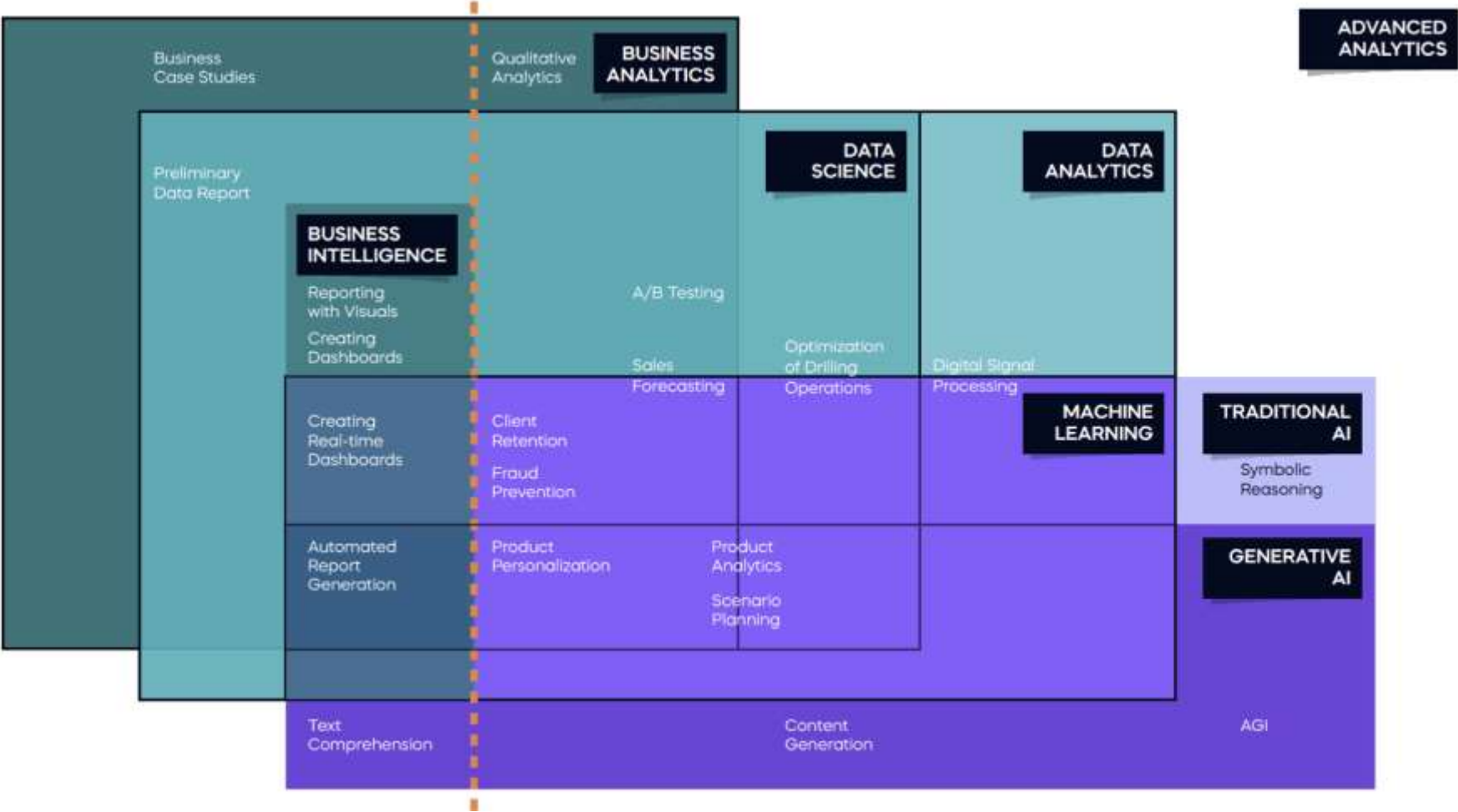
Generating new raw data or content, such as text, code, audio, or images, by relying on patterns from a given training dataset.

*Chat assistants, image generation, code completion.*



# The Data Science Landscape

How the fields overlap, and where everyday activities actually sit.



## KNOWLEDGE CHECK

# Which of the following best describes business intelligence (BI)?

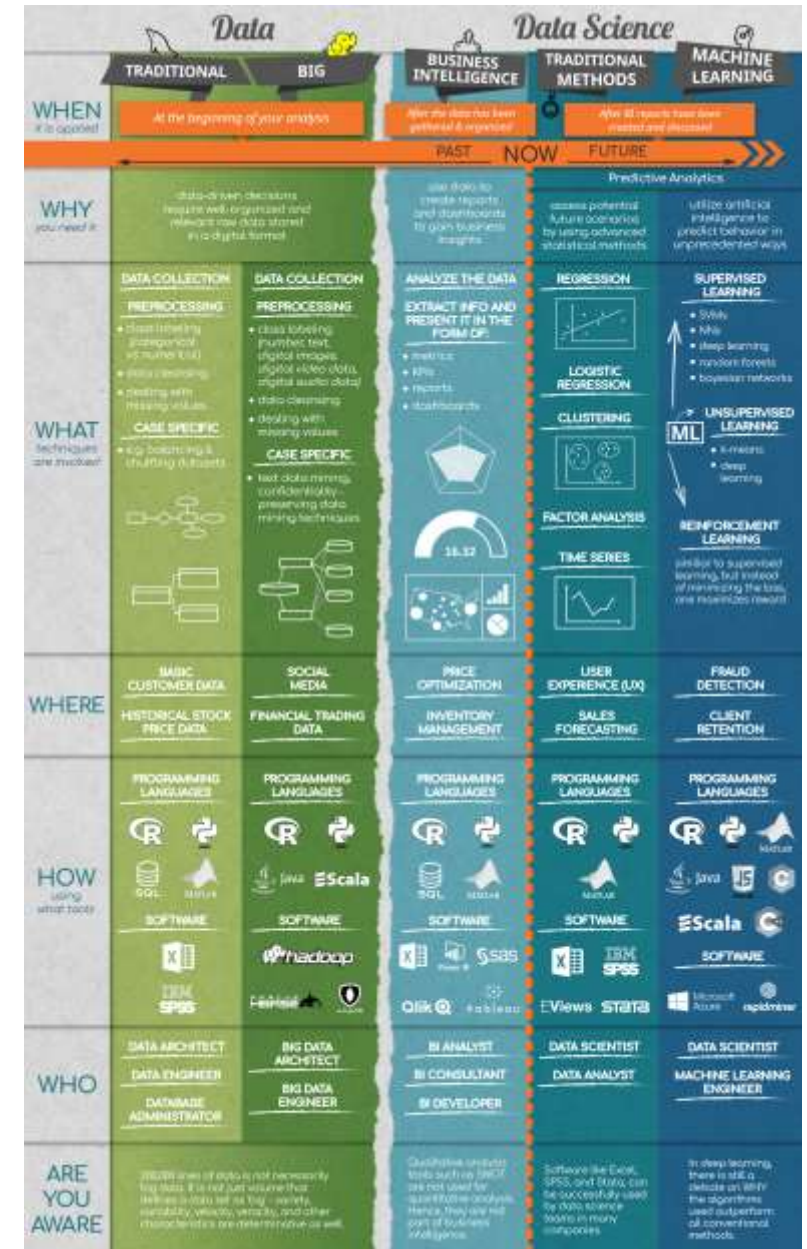
- ☐ A Building models that predict future outcomes from available data
- ☐ B Analysing and reporting historical business data to explain past events
- ☐ C Generating new raw content such as text, code, audio, or images
- ☐ D None of the above

# Data and Data Science at a Glance

One reference chart covering the whole field, column by column: traditional data, big data, business intelligence, traditional methods, and machine learning.

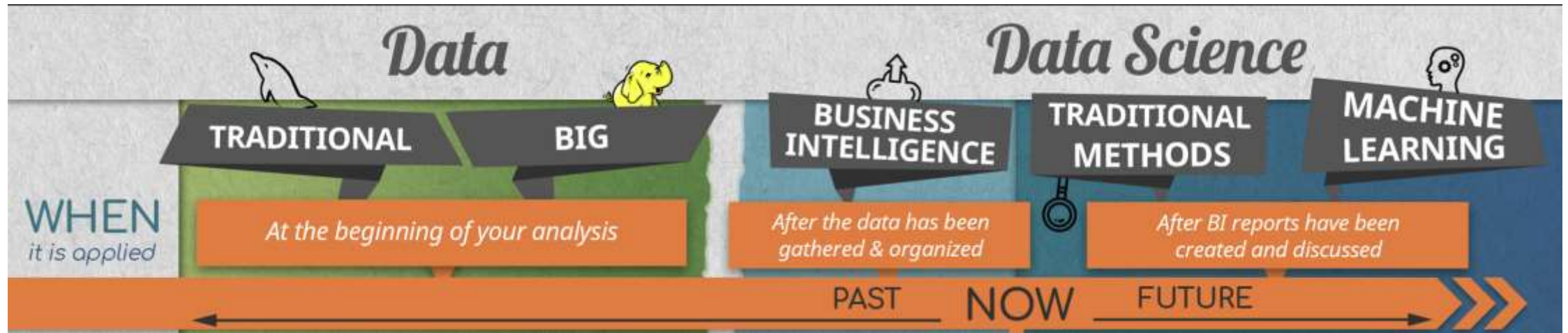
- **WHEN** in the process it is applied
- **WHY** you need it
- **WHAT** techniques are involved
- **WHERE** it gets used in practice
- **HOW** and with which tools
- **WHO** does the work
- **ARE YOU AWARE** the common misconceptions

Source: 365 Data Science



# When Each One Is Applied

The same timeline you have been walking through, laid out end to end.



## Traditional and big data

At the beginning of your analysis.



## Business intelligence

After the data has been gathered and organised.

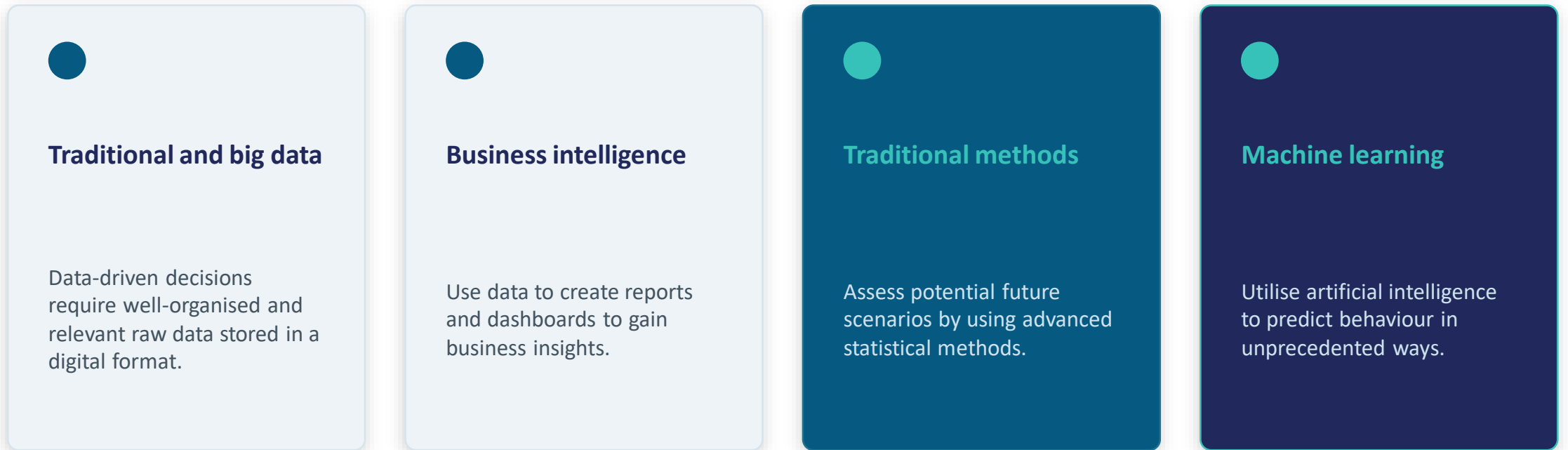


## Traditional methods and ML

After BI reports have been created and discussed.

# Why You Need Each One

Same five columns as the timeline, this time answering what each is for.



Source: 365 Data Science

# Traditional Data

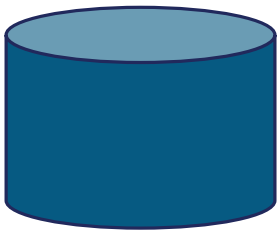
The everyday, orderly kind. Most businesses run on it.

- **Structured**

Rows and columns with a fixed shape you can count on.

- **Managed from one computer**

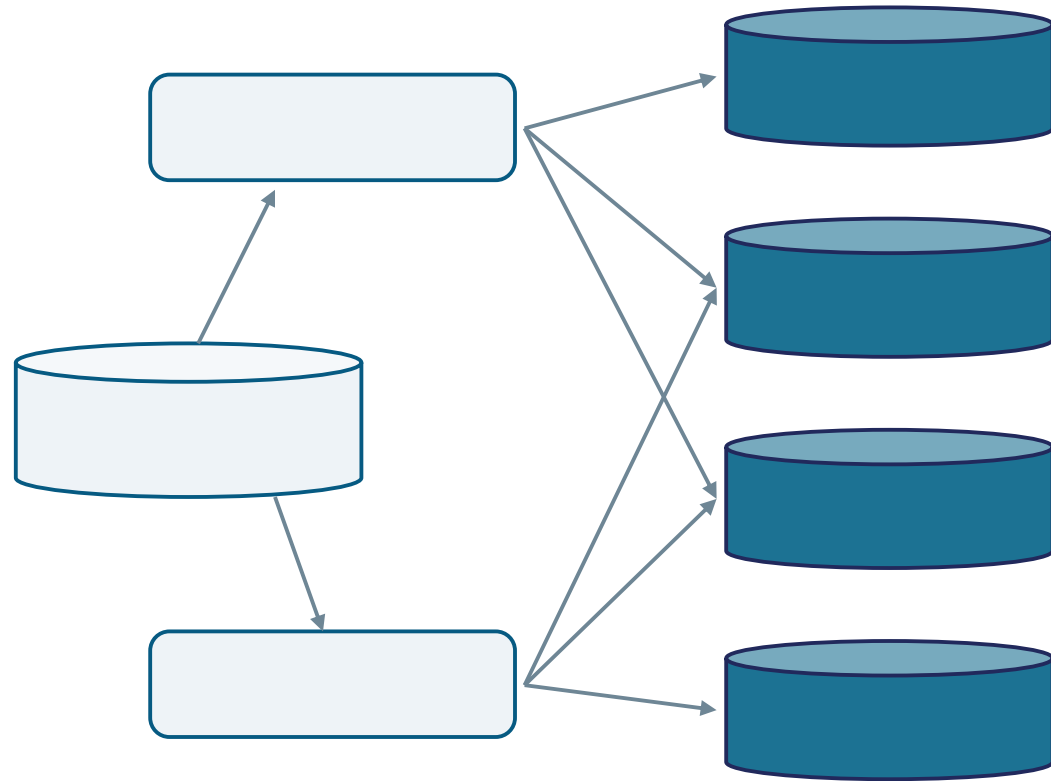
No cluster needed. It fits in a single database on a single machine.



*One machine is enough*

| ID  | Name | Age | ... |
|-----|------|-----|-----|
| 001 | John | 35  | ... |
| 002 | Alan | 22  | ... |
| ... | ...  | ... | ... |

*Structured data sits in a table like this one*



*Stored and processed across many machines*

# Big Data

Too much, too varied, and too fast for a single machine to hold.

## Structured

Tidy rows and columns, the same shape as traditional data.

## Semi-structured

Tags or markers but no fixed table. Think JSON or XML.

## Unstructured

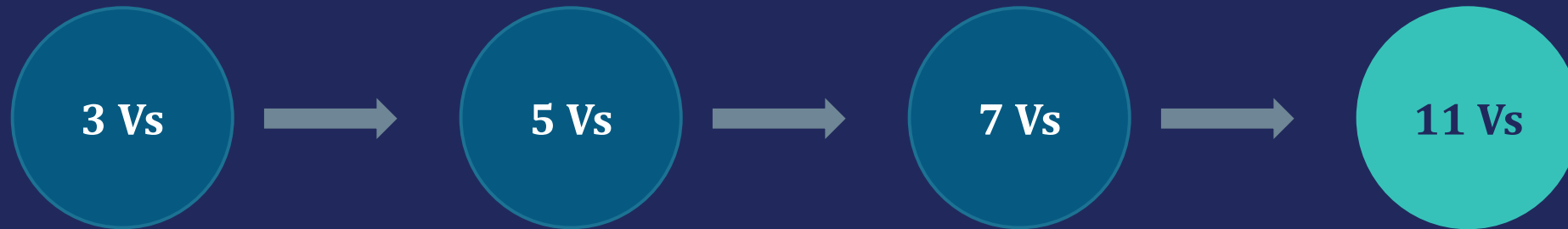
Text, images, audio, video. No inherent shape at all.



# The Vs of Big Data



The list keeps growing as people find new things worth saying about it.



## The original three

**Volume · Variety · Velocity**

*The ones almost everyone agrees on.*

## Added along the way

**Value · Variability · Visualisation · Vision**

*Useful, but the list is not fixed.*

# Traditional Data vs Big Data

The same three questions, answered very differently.

## TRADITIONAL DATA

## BIG DATA

### Volume

Small enough to sit on a single machine.

**Distributed between many computers.**

### Variety

Uniform rows and columns, one shape throughout.

**Text, images, audio, mobile. Many shapes at once.**

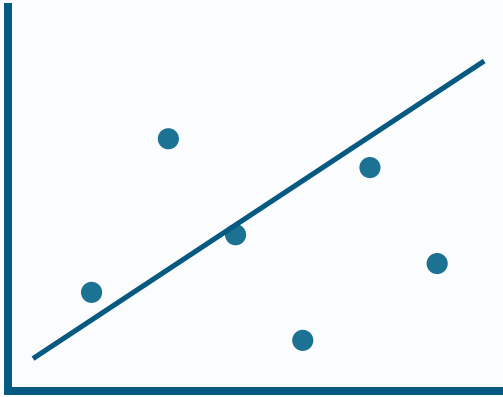
### Velocity

Arrives and updates at a steady, manageable pace.

**Retrieved in real time.**

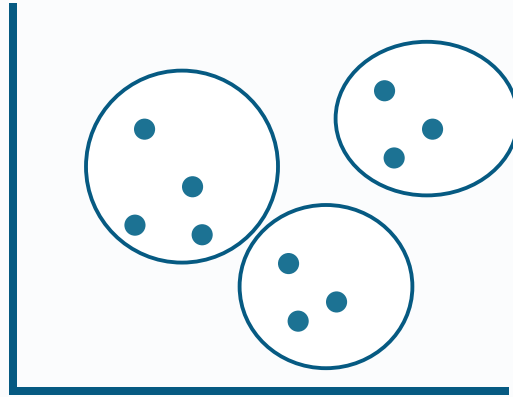
# Traditional Methods

Perfect for forecasting future performance with great accuracy.



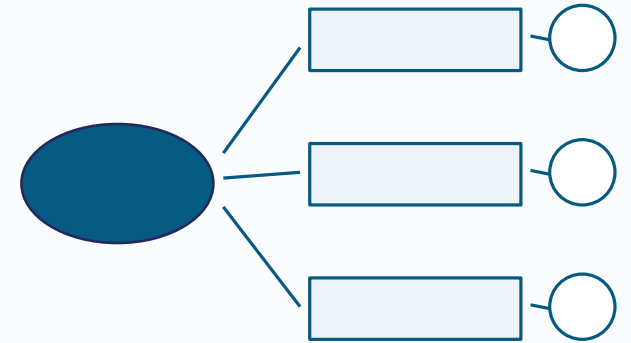
## Regression

Fit a line through the data to estimate one value from another.



## Cluster analysis

Group observations that behave alike, without being told the groups.



## Factor analysis

Reduce many variables down to the few underlying drivers.

## KNOWLEDGE CHECK

**Which column from our infographic contains activities that belong to the field of predictive analytics, and does not aim solely to explain past behaviour?**

**A** Traditional data

**B** Big data

**C** Business intelligence

**D** Traditional methods

END OF PRESENTATION

# Thank you

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Questions are welcome.

